

Assignment 5

Temporal / Sequential Data and Window SQL

A hospital maintains a table of patient vital measurements:

```
Vitals(patient_id, event_time, unit, heart_rate, oxygen_sat)
```

The following tuples are recorded:

patient_id	event_time	unit	heart_rate	oxygen_sat
P100	2026-03-20 08:00:00	ICU	96	92
P100	2026-03-20 08:30:00	ICU	102	89
P100	2026-03-20 09:00:00	ICU	105	87
P101	2026-03-20 08:05:00	Cardiology	88	97
P101	2026-03-20 08:50:00	Cardiology	90	96
P102	2026-03-20 08:10:00	MedSurg	110	94
P102	2026-03-20 08:40:00	MedSurg	112	91
P102	2026-03-20 09:10:00	MedSurg	108	93

a

Fill in the SQL query that uses a window function to return, for each patient, the most recent vital record. Your output should include `patient_id`, `event_time`, `heart_rate`, and `oxygen_sat`.

```
SELECT patient_id, event_time, heart_rate, oxygen_sat
FROM (
    SELECT *,
           ROW_NUMBER() OVER (PARTITION BY patient_id ORDER BY event_time DESC)
    as rn
    FROM Vitals
) t
WHERE rn = 1;
```

b

Fill in the SQL query below that uses a window definition ordered by `event_time` to compute, for each tuple, the running average of `oxygen_sat` for that patient up to and including the current reading. Your output should include `patient_id`, `event_time`, `oxygen_sat`, and `running_avg_oxygen_sat`.

```
SELECT patient_id, event_time, heart_rate, AVG(oxygen_sat)
      OVER (PARTITION BY patient_id ORDER BY event_time DESC) as
running_avg_oxygen_sat
FROM Vitals;
```

C

Fill in the SQL query using a window frame on physical rows to compute, for each tuple, the average `oxygen_sat` over the current reading and the immediately previous reading for the same patient. Your output should include `patient_id`, `event_time`, `oxygen_sat`, and `avg_last_two_readings`.

```
SELECT patient_id, event_time, oxygen_sat,
      AVG(oxygen_sat) OVER (
        PARTITION BY patient_id
        ORDER BY event_time
        ROWS BETWEEN 1 PRECEDING AND CURRENT ROW
      ) as avg_last_two_readings
FROM Vitals;
```

Spatiotemporal Data Model and Query Processing

A hospital system tracks mobile medical devices used in patient care. The data are stored in the following relation:

`DeviceTrack(device_id, time, building, floor, x, y, status)`

where `time` is the timestamp of the observation, `building` and `floor` indicate coarse location, `(x,y)` gives a 2D coordinate on the floor map, and `status` is one of {available, in_use, cleaning, maintenance}.

Suppose the following tuples are recorded:

device_id	time	building	floor	x	y	status
D1	2026-03-20 09:00:00	Main	2	12	30	available
D1	2026-03-20 09:10:00	Main	2	15	34	in_use
D1	2026-03-20 09:20:00	Main	3	8	10	in_use
D2	2026-03-20 09:05:00	Main	2	40	22	cleaning
D2	2026-03-20 09:15:00	Main	2	42	25	available
D3	2026-03-20 09:12:00	North	1	7	18	maintenance

a

Fill in the following query as SQL-like pseudocode: Find the most recent known location (building, floor, x, y) of each device. Returned attributes should include `device_id` and the latest location/ time fields.

Hint: this scenario is similar to Question 1(a).

```
SELECT dt.device_id, dt.time, dt.building, dt.floor, dt.x, dt.y
FROM DeviceTrack dt
JOIN (
    SELECT device_id, MAX(time) AS max_time
    FROM DeviceTrack
    GROUP BY device_id
) AS latest
ON dt.device_id = latest.device_id
AND dt.time = latest.max_time;
```

b

Express the following query in SQL-like pseudocode: Find all devices that were on Main building, floor 2, during the time interval 09:00–09:15, and whose status was available.

```
SELECT DISTINCT device_id
FROM DeviceTrack
WHERE building = 'Main'
    AND floor = 2
    AND status = 'available'
    AND time BETWEEN '2026-03-20 09:00:00' AND '2026-03-20 09:15:00';
```

c

Suppose the hospital now stores millions of such tracking tuples. Briefly explain what kinds of indexing or query-processing support would help answer spatiotemporal queries efficiently. You may discuss ideas such as indexing by time, indexing by location, combining spatial and temporal predicates, partitioning, or windowing.

✓ Answer ✓

We could optimize the structure of the location information by building a quad-tree, which speeds up location searches by reducing the number of nodes required to find one you are looking for.

Similarly, for time, we could utilize a B-tree to optimize searches for timeframes.

Modern Data Quality Issues

Recall the five modern data quality dimensions discussed in class: Validity & Consistency; Completeness & Coverage; Identity & Deduplication; Timeliness & Provenance; Semantic & Annotation Quality.

For each of the following scenarios, annotate it with all applicable data quality dimensions, and briefly explain your reasoning.

a

A nurse note states: "Patient denies chest pain today." But in a training dataset for a clinical text classifier, this note is labeled as `chest_pain = positive`.

✓ **Answer**

This would violate validity and consistency as well as semantic & annotation quality, as the note contradicts the information stored semantically within the `chest_pain` variable.

b

A patient appears in two systems with IDs `P100` and `P100X`. The names are written as "Mary Jones" and "Mary J.", but the date of birth and phone number are the same. It is later confirmed that both records refer to the same person.

✓ **Answer**

This violates Identity & Deduplication, as the same real-world entity is represented twice within the database, leading to inconsistent inferences as some data may not be in scope when querying on one or the other entity.

c

A hospital's retrieval system for policy documents is missing the newest ICU oxygen protocol, so an AI assistant answers a question using an older version of the policy.

✓ **Answer**

This violates Timeliness & Provenance as it does not represent the current real-world situation that is happening, leading to incorrect assumptions being made about the current state of things.

d

A provider directory still lists "Dr. Gomez" as an active attending physician, even though that provider left the hospital network two months ago. A patient-routing workflow still uses that directory.

✓ Answer

This also violates Timeliness & Provenance as well as Validity, as similarly, the database does not represent the current truth.

e

An imaging dataset contains chest X-ray images where several examples labeled “pneumonia” are actually normal scans, and one important patient subgroup is heavily underrepresented.

✓ Answer

This violates Semantic & Annotation Quality as well as Validity, as the data within the database is both incorrect and wrongly labeled.

“Poisoning” Knowledge Graphs

Consider the following RDF-style knowledge graph about a small clinical decision-support setting.

Assume the prefixes `ex:` for the hospital ontology, `pat:` for patients, `prov:` for providers, `cond:` for conditions, and `med:` for medications. The graph contains the following triples:

pat:P100	ex:hasSymptom	cond:LowOxygen
pat:P100	ex:assignedProvider	prov:Lee
pat:P101	ex:hasSymptom	cond:ChestPain
pat:P101	ex:assignedProvider	prov:Patel
prov:Lee	ex:worksIn	ex:ICU
prov:Patel	ex:worksIn	ex:Cardiology
cond:LowOxygen	ex:treatedBy	med:OxygenTherapy
cond:ChestPain	ex:treatedBy	med:ECG

a

Write a SPARQL query that returns all patients who have symptom `cond:LowOxygen` and the provider assigned to them.

```
SELECT ?patient ?provider
WHERE {
  ?patient ex:hasSymptom cond:LowOxygen .
  ?patient ex:assignedProvider ?provider .
}
```

b

Suppose an attacker deletes the triple `pat:P100 ex:hasSymptom cond:LowOxygen` .

Explain how the result of your query in part (a) changes.

✓ **Answer**

It will no longer return patient `P100` nor their respective provider `Lee` .

c

Suppose an attacker inserts the false triple `pat:P101 ex:hasSymptom cond:LowOxygen` .

Explain how the result of your query in part (a) changes.

✓ **Answer**

The query will now additionally return `P101` even though they do not have that condition, along with their provider of `Patel` .

d

Write a SPARQL query that returns all providers working in `ex:ICU` who are assigned to patients with `cond:LowOxygen` .

```
SELECT ?provider
WHERE {
  ?provider ex:worksIn ex:ICU .
  ?patient ex:assignedProvider ?provider .
  ?patient ex:hasSymptom cond:LowOxygen .
}
```

e

Suppose an attacker changes the graph by replacing `prov:Lee ex:worksIn ex:ICU` . with `prov:Lee ex:worksIn ex:Cardiology` .

Explain how the answer to your query in part (d) changes.

✓ **Answer**

The query will now no longer return `Lee` , as the databases does not believe he works in the ICU, returning an empty set.